Laboratory 6 – Pop Machine

PLC – Sequential Functional Chart Program

Description

You are building the latest soda pop machine design. The following is a description of the system's operation.

Procedure

- Create the program
- Demonstrate its operation
- Document your work in a lab report.

Inputs

Switch_1 – Coin is inserted
Switch_2 – Repeat switch (start over switch)
Switch_3 – Add Ice to drink
Switch_4, Switch_5 = {00 = Orange, 01 = Peach, 10 = Grape, 11 = Lemon}
Switch_6 – (optional) Stop flashing LEDs in Make selection step

Outputs

LED_1 – Local:2:O:data.0 – see procedure
LED_2 – Local:2:O:data.1 – see procedure

120V Socket #1 – Local:2:O:data.2 – Music
120V Socket #2 – Local:2:O:data.3 – Pop Mixer

Cup = {'Orange', 'Peach', 'Grape', 'Lemon'} {' + '} {'Ice'}

Operation

Initial Step

- Wait until coin is inserted (push button switch 1)
- All LEDs are off
- All 120 V Sockets are off

Make_Selection Step 001 (make your selection)

- User makes selection: Switch_4, Switch_5 = {00 = Orange, 01 = Peach, 10 = Grape, 11 = Lemon}
- Turn on running LED (#1) and leave it on until step 009
- Flash starting warning LEDs (#2) at a rate of 1 Hz. (Warning machine starting LED)
    - And continue flashing LED (#2) for 8 seconds

    - Optionally continue flashing until Switch6 is press (Enter switch)
- Leave this state after 3 seconds

Music_On Step 002

- Turn on music (120 V Socket #1)

Mixer_On Step 003

- Turn on the pop mixer (120 V Socket #2)
- Wait 2 seconds
- Turn off mixer

Drop_Cup Step 004

- Drop the cup (i.e., set Cup := ‘’;)
- Wait 1 second
- Go to both Step 006 & 007

Dispense_Pop Step 006

- Based on Switch 4&5 = {00 = Orange, 01 = Peach, 10 = Grape, 11 = Lemon})
- Set pop_type := {‘Orange’, ‘Peach’, ‘Grape’, ‘Lemon’};
- Cup := CONCAT(Cup,’ + ‘,pop_type);
- Wait for 2 seconds
- Goto Step 008

Add_Ice Step 007 (Add ice if wanted – switch #3)

- If wanted (i.e., switch #3 = 1),
    - Cup := CONCAT(Cup,‘ + ‘, Ice);
    - wait 3 seconds
- Goto Step 008

Wait_Step 008

- Make sure both Steps 006 and 007 are done

Finished Step 009 (Finished)

- Go wild with the LEDs
- Turn off the Music after the LEDs go wild
- Wait for Repeat switch #2
- Go to Step 000

Optional: Try using INSERT() instead of CONCAT().

Appendix RSLogix 5000 v20 String Manuipation

Literal strings are not supported in RSLogix 5000 v20. Instead use strings tags that have an initial value set in Monitor Tags.

To initialize string tags

1. In Monitor Tags in the value field of a Tag of type STRING, click '…' to open a dialog box and then enter the string:
   For example:
   OrangeSTR initialize as 'Orange'
   IceSTR initialize as ' + Ice.'
   NullSTR initialize as ''.

To get a clean Cup:

CONCAT(NullSTR,NullSTR,Cup);

To add Orange to the cup:

CONCAT(Cup,OrangeSTR,Cup);

To add Ice to the cup:

CONCAT(Cup,IceSTR,Cup);

To see the cup

1. Select Quick Watch (F3).
2. Select Cup the tag to watch.